

ENERGY AND ENVIRONMENTAL HAZARD

Oil & Natural Gas Wells

An operating, idle, plugged, or orphaned well can create different air, soil, subsurface, fire, and land-use concerns.



1 Understand the factor

What it is - and why it matters

Oil and natural-gas wells may be actively producing, temporarily idle, undergoing maintenance, plugged and abandoned, or orphaned without a financially responsible operator. Associated equipment can include tanks, separators, compressors, pumps, gathering lines, injection wells, and access roads. Potential hazards include methane, volatile organic compounds, hydrogen sulfide, spills, blowouts, fire, noise, truck traffic, subsurface migration, and land subsidence. A well symbol on a map does not reveal its construction, status, integrity, emissions, or whether old facilities remain below grade.

How risk can reach a household

- Methane or other gases can migrate through defective casing, abandoned boreholes, soil, utility trenches, or nearby structures.
- Produced water, oil, drilling fluids, and tank releases can affect soil, surface water, or groundwater.
- Compressors, pumps, flares, engines, and truck loading can emit pollutants and create noise, light, and traffic impacts.
- Well workovers, pressure events, tank fires, and blowouts can create short-term emergency zones beyond the lease boundary.

What people often miss

- A well labeled 'plugged' may still require records review because plugging standards and documentation changed over time.
- Old wells can be hidden beneath landscaping, roads, homes, or later development, especially in historic production areas.
- Distance is only a screening measure; wind, terrain, operations, equipment, production volume, and integrity are more informative.
- Mineral rights and surface rights may be separated, allowing access or future operations subject to recorded agreements and law.

? Five checks before buying

1

Use the state oil-and-gas well database to identify all mapped wells, status, depth, operator, inspections, violations, plugging records, and nearby facilities.

2

Review title for mineral reservations, surface-use agreements, access roads, pipelines, easements, indemnities, and restoration obligations.

3

Inspect for tanks, vents, stained soil, petroleum odor, dead vegetation, subsidence, abandoned pads, and unusual utility penetrations.

4

For a home over or near a legacy well, consult qualified environmental and geotechnical professionals about methane and soil-gas screening.

5

Ask the insurer and lender about active industrial use, pollution exclusions, well-related subsidence, fire protection, and prior claims.

IMPORTANT DISTINCTION Map proximity is a screening signal - not proof of exposure, damage, or insurability. Confirm the exact feature, current condition, pathway, and property-specific controls.

HEALTH, PROPERTY AND COVERAGE

How this factor can affect a household

FACTOR 09

H Health impacts

- Acute high exposures can cause headache, dizziness, nausea, eye or airway irritation, loss of consciousness, burns, or asphyxiation.
- Long-term health evidence near oil and gas development varies by pollutant, operation, distance, and study design; confirmed exposure data matter more than proximity alone.
- Hydrogen sulfide is highly toxic at elevated concentrations; methane mainly presents explosion and oxygen-displacement hazards.
- Noise, odors, bright lights, and heavy truck activity can affect sleep, stress, outdoor activity, and quality of life.

P Property impacts

- Spills and leaking equipment can require soil removal, groundwater investigation, vapor controls, and long-term agency oversight.
- Access agreements, mineral rights, industrial traffic, noise, and visual impacts can reduce usable land or affect resale perception.
- Methane intrusion, subsidence, abandoned piping, and inaccessible wellheads can complicate foundations, additions, pools, and redevelopment.
- A nearby emergency can require evacuation or sheltering and interrupt roads, utilities, schools, and neighborhood access.

I Insurance implications

- Homeowners insurance commonly excludes pollution, gradual leakage, land, settling, and business or industrial liabilities.
- A responsible operator may have legal duties, yet insolvency, orphan status, causation disputes, and cleanup timelines can leave owners exposed.
- Sudden fire or explosion may be covered, but remediation of pre-existing contamination or an abandoned well may not be.
- Specialty environmental, builder, commercial, or contractual coverage may be relevant for redevelopment but is not a substitute for due diligence.

Possible long-term health implications of buying nearby

- Living near an intact, compliant well does not automatically mean harmful exposure; the health question depends on measured emissions and a pathway into air, soil, or water.
- Private-well users may need targeted water testing when there is a credible release or agency advisory.
- Persistent odors or symptoms that track operations deserve documentation and reporting, but odor alone does not identify concentration or medical cause.
- Stress may persist where operations are noisy, complaints recur, or responsibility for old wells is uncertain.

Evidence lens

Oil-and-gas hazards are real, but health conclusions for a particular home require information about the well's status, integrity, emissions, pollutants, exposure pathway, and duration.

Plan more carefully for

Children

Older adults

Pregnancy

Heart/lung conditions

Mobility or power needs

Insurance reality: Availability, eligibility, exclusions, deductibles, and limits vary by state, carrier, property, and cause of loss. Get an address-specific written quote before the purchase contingency expires.

PRACTICAL RISK REDUCTION

Precautions and a real-life example

FACTOR 09

V Verify the actual risk

- Download the complete well file, plugging report, inspection history, spill record, and operator information from the state regulator.
- Confirm mineral ownership and obtain professional review when a well is on, under, or immediately adjacent to the parcel.

P Protect the property

- Keep wellheads and access areas visible and protected; never cap, cover, excavate, or build over a well without regulatory approval.
- Maintain drainage, prohibit ignition sources near equipment, and promptly report leaks, damaged fencing, or illegal dumping.

M Monitor and respond

- Use listed gas alarms where recommended, document odors and operating conditions, and follow agency instructions for sampling.
- Leave and call 911 for strong gas odor, hissing, visible release, fire, sudden illness, or a methane-alarm event.

I Prepare insurance records

- Review pollution, explosion, subsidence, earth-movement, business-use, and additional-living-expense wording.
- Keep title documents, baseline photographs, sampling reports, and agency correspondence for future claims and resale.

REAL-LIFE EXAMPLE

Aliso Canyon Natural Gas Storage Leak

October 2015-February 2016

A well failure at the Aliso Canyon underground natural-gas storage facility near Porter Ranch, California, released natural gas for months. Residents reported odors and symptoms, thousands of households relocated, and regulators conducted emergency response, investigation, and later oversight of the facility.

Why it matters: The event demonstrated that well integrity, subsurface storage pressure, odorant, monitoring, and communication can affect communities beyond a facility boundary for an extended period.

[Open official incident record - cpuc.ca.gov >](https://cpuc.ca.gov)

Questions to resolve before making a decision

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2. Review title for mineral reservations, surface-use agreements, access roads, pipelines, easements, indemnities, and restoration obligations.
3. Inspect for tanks, vents, stained soil, petroleum odor, dead vegetation, subsidence, abandoned pads, and unusual utility penetrations.
4. For a home over or near a legacy well, consult qualified environmental and geotechnical professionals about methane and soil-gas screening.

Official sources and mapping tools

[California WellFinder - conservation.ca.gov >](https://conservation.ca.gov)[EPA oil and natural-gas air rules - epa.gov >](https://epa.gov)[PHMSA underground natural-gas storage - phmsa.dot.gov >](https://phmsa.dot.gov)